Name:Abhishek Yadav.USN No:CS23071.ML lab-10 Aim:To perform hierarchical clustering using agglomerative methods and interpret dendrograms for cluster formation.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import scipy.cluster.hierarchy as sch
from sklearn.cluster import AgglomerativeClustering
import seaborn as sns
from scipy.cluster.hierarchy import dendrogram,linkage
from sklearn.preprocessing import StandardScaler
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
path = "/content/drive/MyDrive/dataset_10/Mall_Customers.csv"
df = pd.read_csv(path)
df.head(5)
```

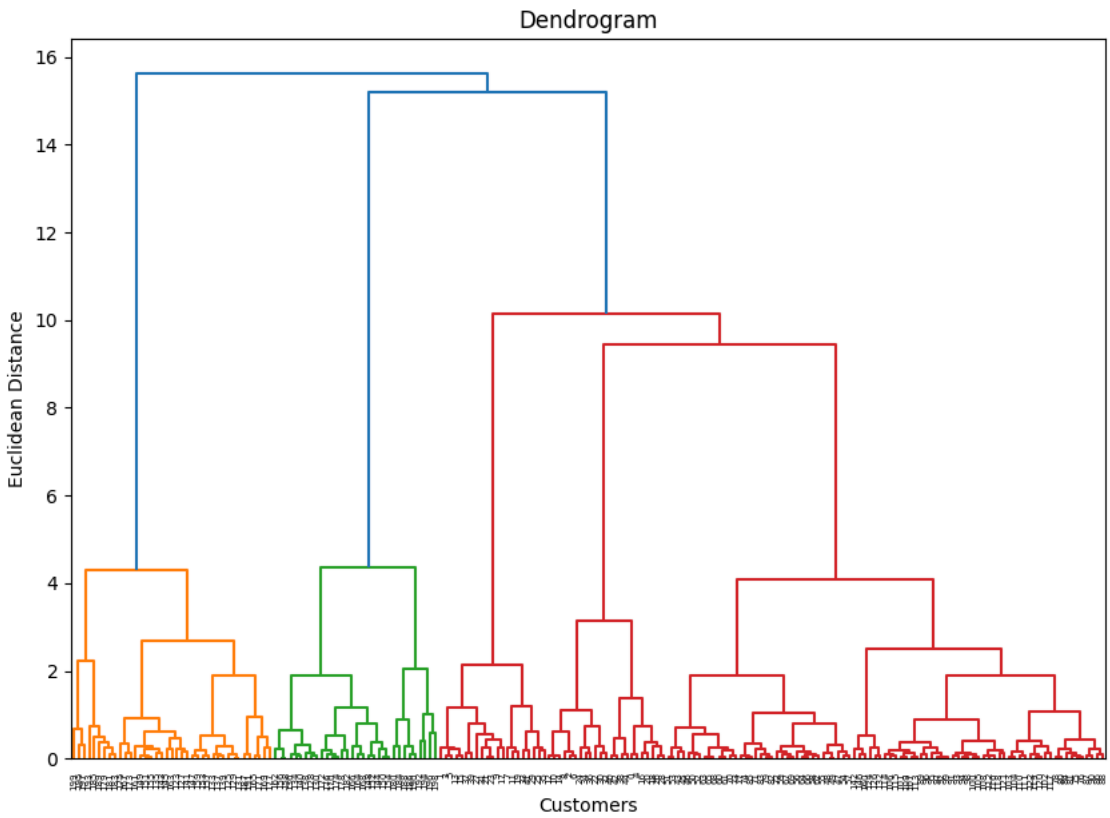
	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Next steps: Generate code with df New interactive sheet

```
x = df[['Annual Income (k$)','Spending Score (1-100)']].values
```

```
scaler=StandardScaler()
x_scaled=scaler.fit_transform(x)
```

```
plt.figure(figsize=(10,7))
dendrogram(linkage(x_scaled,method='ward'))
plt.title('Dendrogram')
plt.xlabel('Customers')
plt.ylabel('Euclidean Distance')
plt.show()
```

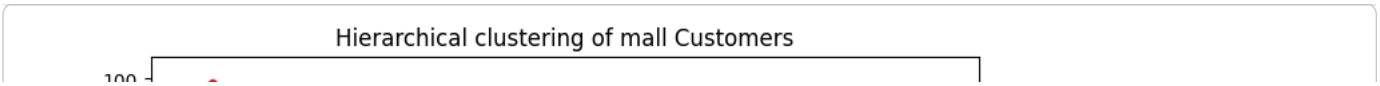



```
hc = AgglomerativeClustering(n_clusters=5,linkage='ward')
cluster = hc.fit_predict(x_scaled)
```

```
df['cluster'] = cluster
print(df.head(5))
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)  cluster
0           1    Male   19                  15                      39        4
1           2    Male   21                  15                      81        3
2           3  Female   20                  16                       6        4
3           4  Female   23                  16                      77        3
4           5  Female   31                  17                      40        4
```

```
plt.figure(figsize=(8,6))
sns.scatterplot(x='Annual Income (k$)', y='Spending Score (1-100)',hue='cluster',data=df, palette='tab10')
plt.title('Hierarchical clustering of mall Customers')
plt.show()
```


Hierarchical clustering of mall Customers
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